



DriverJinn: Topology-Aware Deep Learning for Cancer Driver Gene Identification in Gene Interaction Networks

Authors: Siddharth Rajesh, Xander Krohannon Department of Biomedical Engineering and Informatics
Luddy School of Informatics, Computing, and Engineering

Abstract

- Driver genes drive progress tumor growth in the body by being the first genes to mutate.
- Identification of these genes continue to be a difficulty owing to the problem of **low signal-to-noise ratio**.
- DriverJinn proposes a novel implementation of ranking entire gene lists curated from a **curvature-aware GNN** trained on graph data.
- Evaluated across three biological networks (GGNet, PathNet, PPNet), DriverJinn achieves **AUROC ~0.81** and **NDCG@50 ~0.52**.
- Identified a **chromatin remodelling cluster of 7 genes** that are the center of a densely connected functional module

Objective

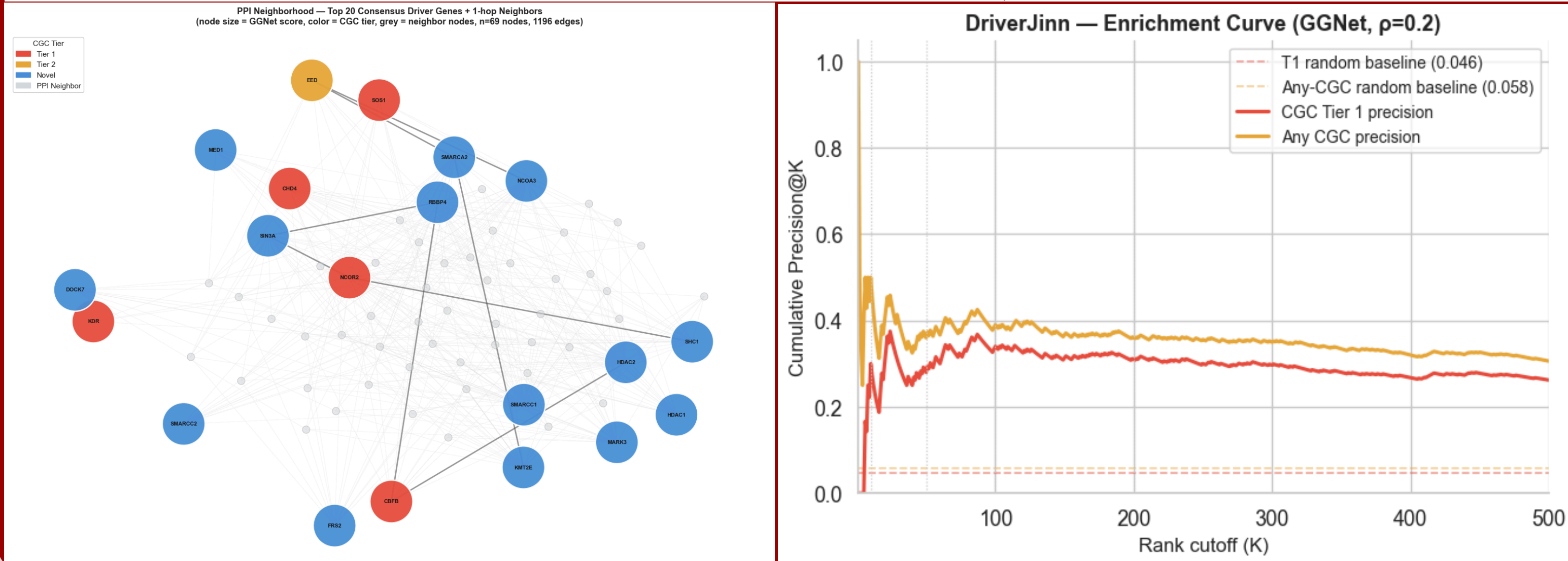
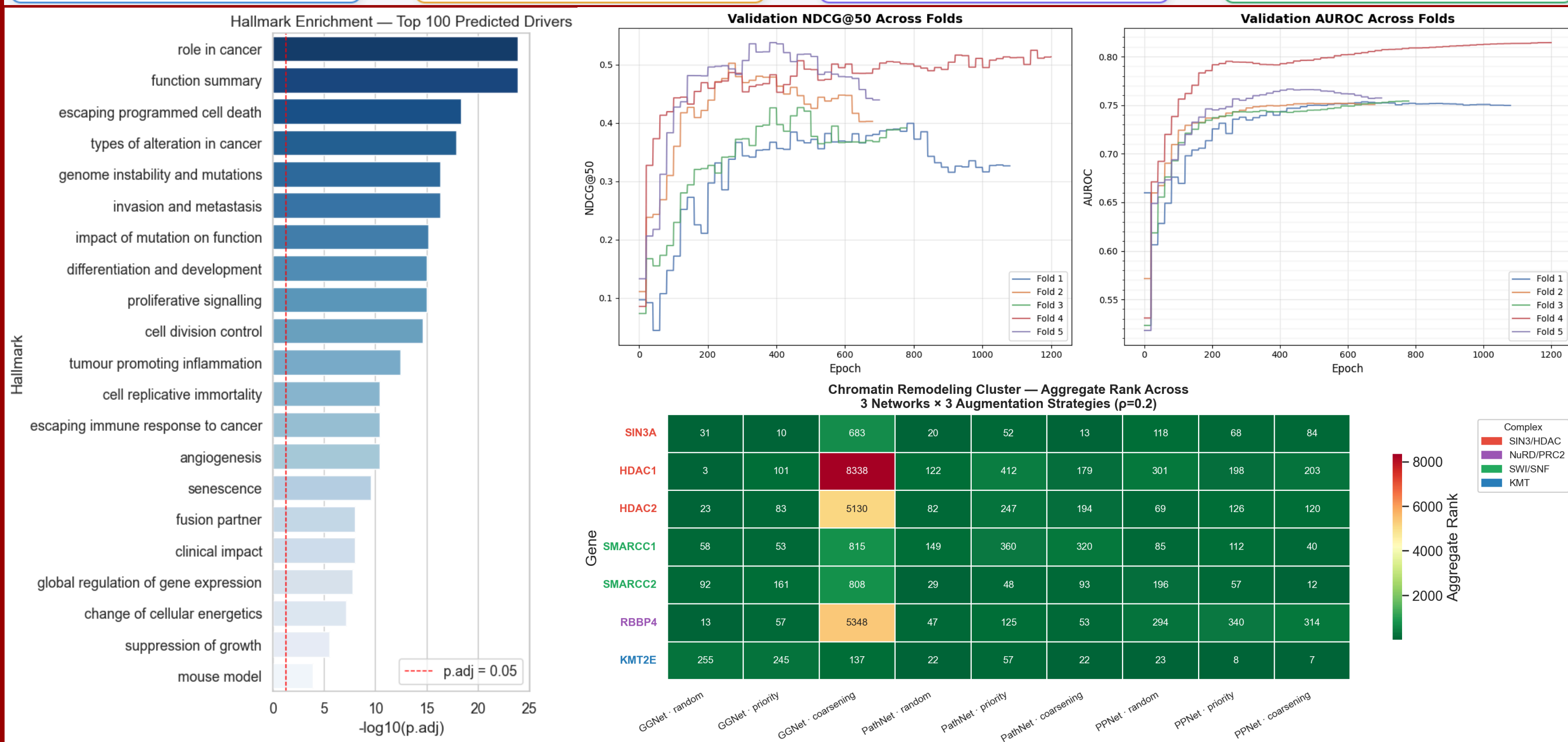
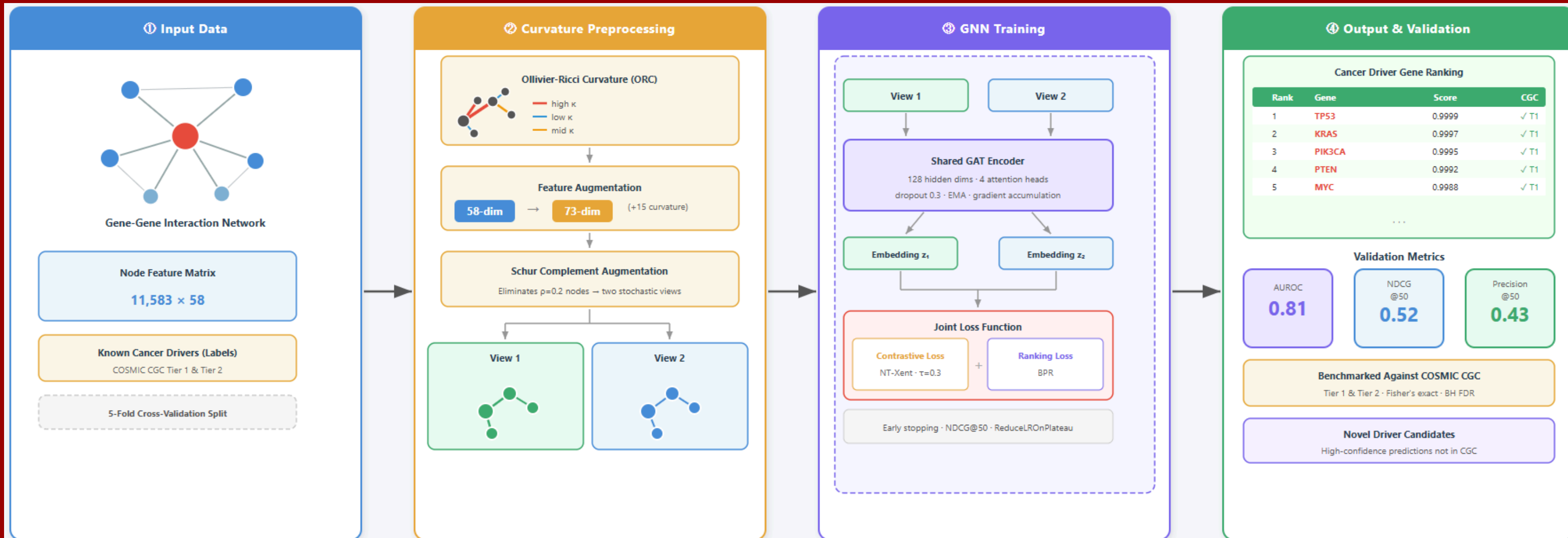
Current driver gene predictors are often network-specific and do not leverage the complete network topology. DriverJinn addresses this by..

- Rank cancer driver genes** across genome-wide gene lists using curvature-aware graph representations from gene interaction networks.
- Improve robustness** by training with contrastive learning and Schur complement augmentation to reduce reliance on any single graph topology.
- Identify cross-network consensus drivers** - genes consistently predicted as drivers across heterogenous biological networks.

Results

- Fig 1: **Hallmark Enrichment bar chart** showing 21 over-represented COSMIC cancer hallmarks
- Fig 2a) **Validation NDCG@50** (Ranking metric)
- Fig 2b) **Validation AUROC** training curves
- Fig 3: **Cross Network x Elimination Strategy Rank Heatmap** to compare the rankings of the chromatin accessibility genes
- Fig 4: **PPI subgraph** of the top 20 consensus driver genes predicted by DriverJinn plus their "bridge neighbors"
- Fig 5: **Enrichment curves** - DriverJinn enriches known drivers 8-10× above random in the top-ranked genes

Methodology and Results



Conclusion

- The metrics demonstrate that curvature-aware graph topology improves consensus ranking.
- The top 100 predictions are **significantly enriched across all 21 cancer hallmarks**, confirming the model recovers biologically coherent driver candidates rather than statistical artifacts.
- A functionally coherent cluster of 7 chromatin remodelling genes (**SIN3A, HDAC1, HDAC2, SMARCC1, SMARCC2, RBBP4, KMT2E**) was consistently ranked among the top predictions with near-maximal confidence across all folds and **8 of 9** network-strategy conditions, despite none appearing in the CGC curated census — representing the strongest novel finding of this study
- Cross-network consensus ranking improves **prediction robustness** and distinguishes **high-confidence novel candidates** from network-architecture-dependent predictions.

Future Directions

- Experimental validation of the chromatin remodeling cluster, particularly **SIN3A** and **RBBP4**, as candidate cancer drivers in cancer cell line perturbation studies
- Extend the curvature preprocessing to incorporate **dynamic curvature changes** under different cancer subtypes, enabling tumor-type-specific driver prediction
- Scale evaluation to **pan-cancer settings across TCGA cohorts** to assess whether the chromatin remodeling predictions are cancer-type-specific or universal
- Study how network topology choice interacts with driver gene hub structure using the coarsening augmentation method that eliminates high-degree hub nodes.

References



GitHub

